Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

The dataset was cleaned and analyzed using Python (Pandas). The following steps were performed:

- Loaded dataset using `read_csv()`
- Checked dataset structure using:
  a. `df.shape`
  b. `df.columns`
- Generated statistical summary using:
  a. `df.describe()`
- Checked data types using:
  a. `df.info()`
- Identified missing values using:
  a. `df.isnull().sum()`

Missing values in the **Review Rating** column were handled appropriately.

```python
from pydoc import describe
import pandas as pd
df=pd.read_csv( filepath_or_buffer: r"C:\Users\dhana\Downloads\customer_shopping_behavior.csv",sep='\t')#if u want to select a
#particular sheet ,sheet_name=sheet1)

#print("Step 1: First 5 Rows")
print(df.head())

#print("Step 2: Last 5 Rows")
print(df.tail())
```

```
C:\Users\dhana\python.exe C:\Users\dhana\PycharmProjects\official_project\first_project.py
   Customer ID  Age  ... Payment Method Frequency of Purchases
0            1   55  ...          Venmo            Fortnightly
1            2   19  ...           Cash            Fortnightly
2            3   50  ...    Credit Card                 Weekly
3            4   21  ...         PayPal                 Weekly
4            5   45  ...         PayPal               Annually

[5 rows x 18 columns]
      Customer ID  Age  ... Payment Method Frequency of Purchases
3895         3896   40  ...          Venmo                 Weekly
3896         3897   52  ...  Bank Transfer              Bi-Weekly
3897         3898   46  ...          Venmo              Quarterly
3898         3899   44  ...          Venmo                 Weekly
3899         3900   52  ...          Venmo              Quarterly
```

```python
"""
df.drop( labels: "promo_code_used",axis=1,inplace=True)
from sqlalchemy import create_engine
engine=create_engine("mysql+pymysql://root:Dhana123#@Localhost:3306/customer")
df.to_sql( name: 'customer',con=engine,index=False,if_exists='replace')
#con= connect to this engine we created earlier
#why not pymysql because we want to do verything from create to insert manually
#here it is automatical with in two steps
```

```
7   27     young
8   26     young
9   57     adult
<ArrowStringArray>
[   'Fortnightly',         'Weekly',       'Annually',      'Quarterly',
      'Bi-Weekly',        'Monthly', 'Every 3 Months']
Length: 7, dtype: str
  frequency_of_purchases  count
0         Every 3 Months    584
1               Annually    572
2              Quarterly    563
3                Monthly    553
4              Bi-Weekly    547
5            Fortnightly    542
6                 Weekly    539

Process finished with exit code 0
```

```python
print(df.shape)
```

```
[5 rows x 18 columns]
Index(['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category',
       'Purchase Amount (USD)', 'Location', 'Size', 'Color', 'Season',
       'Review Rating', 'Subscription Status', 'Shipping Type',
       'Discount Applied', 'Promo Code Used', 'Previous Purchases',
       'Payment Method', 'Frequency of Purchases'],
      dtype='str')
(3900, 18)
        Customer ID          Age  ... Payment Method Frequency of Purchases
count   3900.000000  3900.000000  ...            3900                   3900
unique          NaN          NaN  ...               6                      7
top             NaN          NaN  ...          PayPal         Every 3 Months
freq            NaN          NaN  ...             677                    584
mean    1950.500000    44.068462  ...             NaN                    NaN
std     1125.977353    15.207589  ...             NaN                    NaN
min        1.000000    18.000000  ...             NaN                    NaN
25%      975.750000    31.000000  ...             NaN                    NaN
50%     1950.500000    44.000000  ...             NaN                    NaN
75%     2925.250000    57.000000  ...             NaN                    NaN
max     3900.000000    70.000000  ...             NaN                    NaN

[11 rows x 18 columns]
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
```

5. Data Storage (MySQL)

After cleaning, the dataset was transferred into MySQL database for further analysis.

- Created database and table
- Imported cleaned dataset into MySQL using python(sqlalchemy)
- Verified data using sample query

6.Data Analysis using SQL (Business Transactions)

I have performed structured analysis in MySQL to answer key business questions:

Revenue by Gender

Analyzed total revenue contribution by each gender.

Average Purchase Amount

Calculated the average spending behavior of customers.

Discount Impact Analysis

Identified customers making purchases above average with discounts.

Top Products by Rating

Found top-performing products based on customer reviews

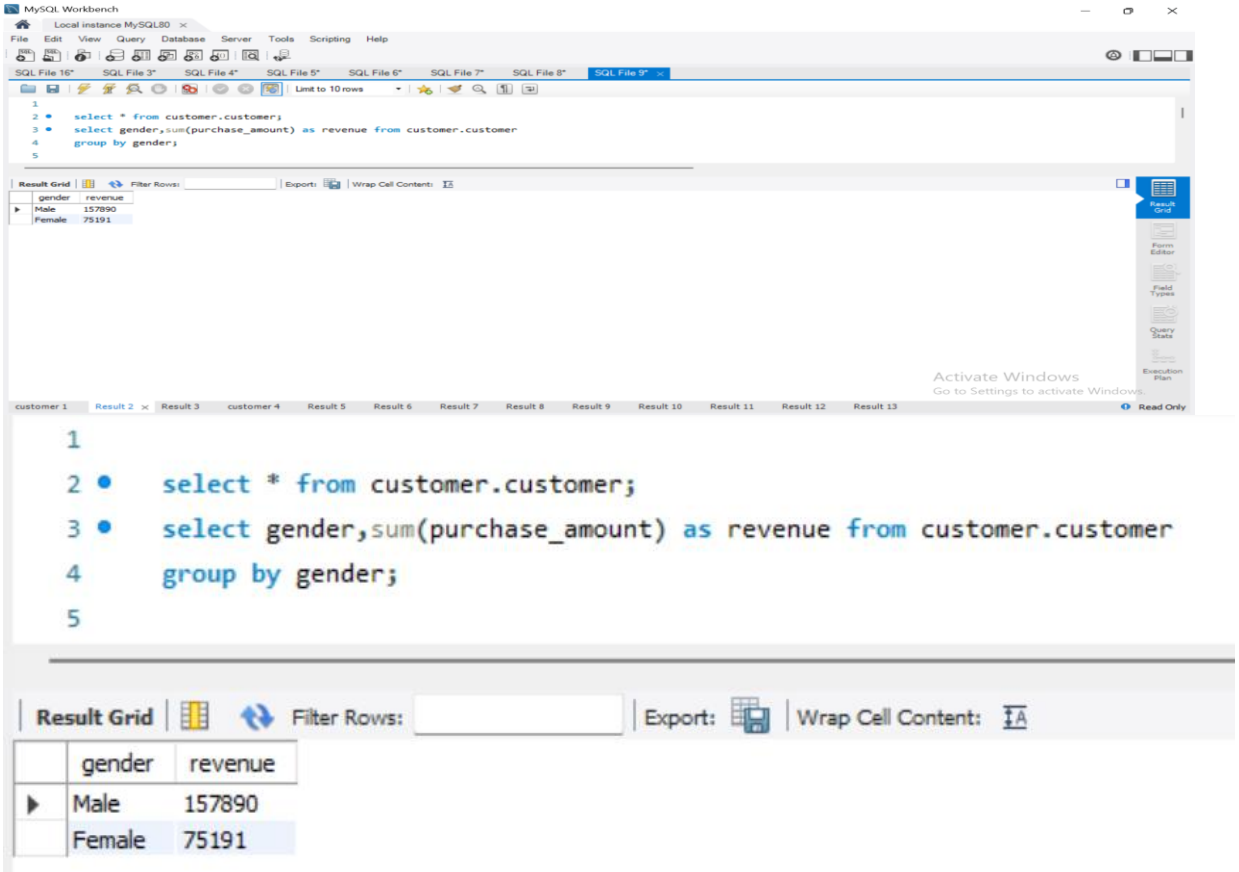
Subscription Analysis

Compared revenue and order count between subscribed and non-subscribed users.

Customer Segmentation

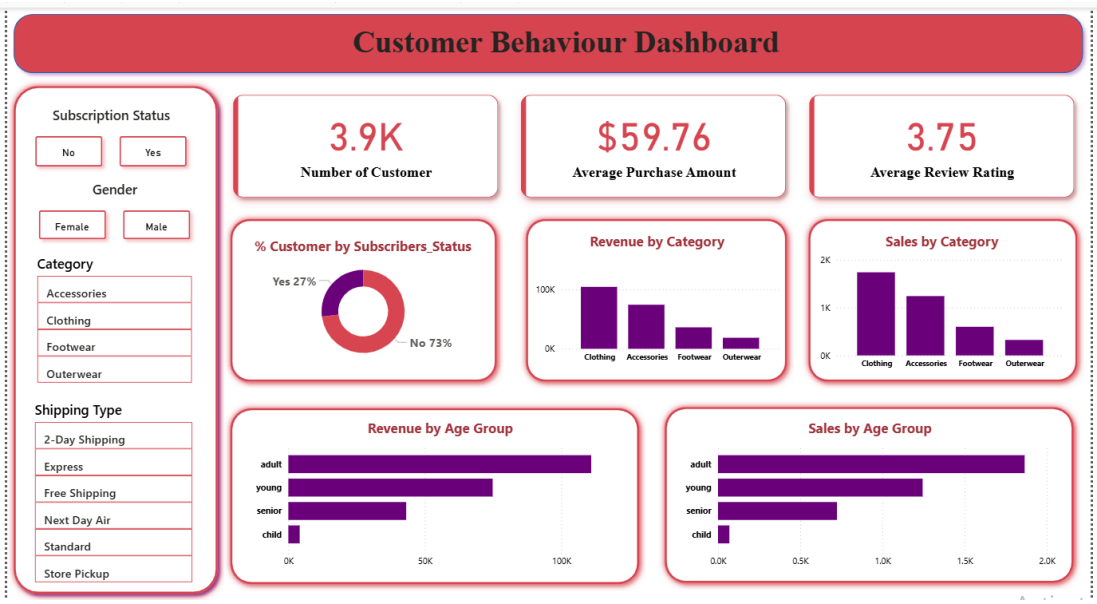
Segmented customers into:

1) Loyal
2) Returning
3) New



7.Dashboard in Power BI

Finally, I built an interactive dashboard in **Power BI** to present insights visually.



8.Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.